   B. Note that the domain for this function is *[0, 2]^2*.
2. Declare the corresponding test data with `xs` for a two-dimensional grid representing the domain and `ys` for the function values of `xs`.
3. Modify the helper function that visualizes optimization progress. For one-dimensional objective functions, it's easy to visualize the GP predictions along with the acquisition scores. For this two-dimensional objective, the helper function should generate a plot of two panels: ne showing the ground truth and the other showing the acquisition scores. Both panels should also show the labeled data. The plot should look similar to Figure 4.18.

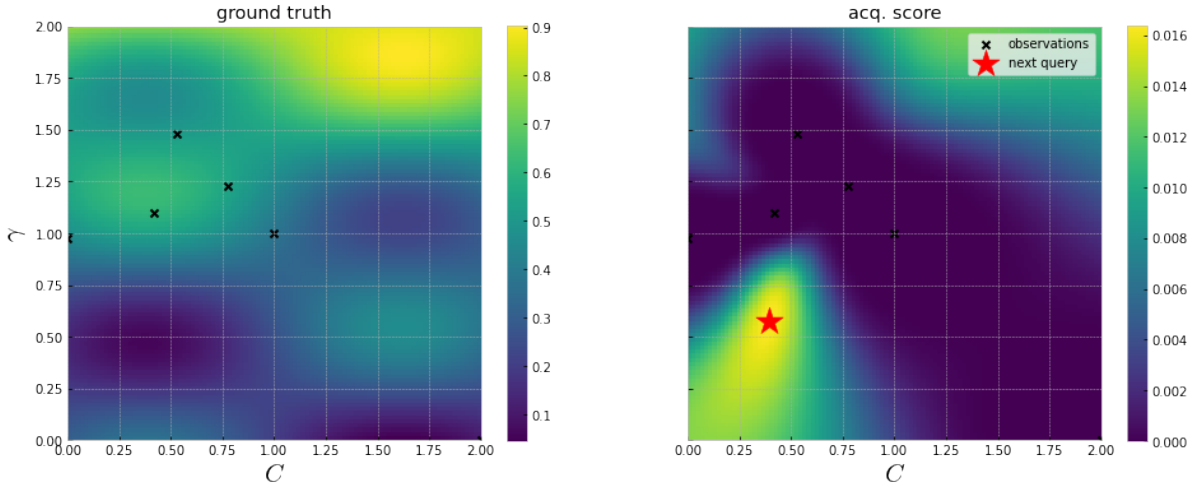


Figure 4.18 A reference of what the helper function that visualizes BayesOpt progress should look like The left panel shows the true objective function, while the right panel shows the acquisition scores.

4. Copy the GP class from Chapter 3's exercise, which implements a Matérn 2.5 kernel with ARD. Further modify this class to make it integratable with BoTorch.
5. Reuse helper function `fit_gp_model()` and the for loop that implements BayesOpt.
   A. The initial training data set should contain the point in the middle of the domain: (1, 1).
   B. As our search space is two-dimensional, make the search for the point maximizing the acquisition score more exhaustive by setting `num_restarts = 40` and `raw_samples = 100` in `botorch.optim.optimize_acqf()`.
   C. Set the number of queries that we can make (the number of times we can evaluate the objective function) to 20.
6. Run the Probability of Improvement (PoI) policy on this objective function. Observe that the policy once again gets stuck at a local optimum.
7. Run the modified version of PoI where the minimum improvement threshold is set at 0.1.
   A. See Exercise 1 for more detail about setting the minimum improvement threshold for PoI.
   B. Observe that this modification again leads to better optimization performance.
   C. What's the first iteration where we reach an accuracy of at least 90 percent? What are the model's parameters achieving this accuracy?
8. Run the Expected Improvement policy on this objective function.
   A. Observe that the policy outperforms PoI.
   B. What's the first iteration where we reach an accuracy of at least 90 percent? What are the model's parameters achieving this accuracy?
9. Inspecting the performance of a policy based on a single run of BayesOpt might be misleading. It's better for a BayesOpt experiment to be repeated multiple times with

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